

Statistics Exercises - Confidence Intervals and Hypothesis Testing

1 Confidence Intervals

Exercise 1. (Dublin Tourism Spending)

The Dublin Tourism Commission is interested in the average amount of money a typical university student spends per day during the Easter holidays. They survey 35 students and find that the mean spending is 63.57 euros with a population standard deviation of 17.32 euros.

1. Develop a 95% confidence interval for the population mean daily spending.
2. Explain what this implies.
3. What level of confidence is associated with an interval of €58.62 to €68.52 for the population mean daily spending?
4. Is there evidence that the average amount spent by students is less than €70?

Exercise 2. (Book Publishing Prices)

A publishing company is interested in publishing new hard back copies of Phillip Pullman's trilogy His Dark Materials in a 3-book boxed set. A recent graduate is asked to randomly select 40 book shops in Dublin and find the sales price of the boxed set for 2018. The mean price in 2018 across the 40 book stores was €27. Assume that the population standard deviation is €5.

1. Using the information from the sample calculate a 99% confidence interval for the population mean price in 2018
2. What did we need to assume about the population distribution and why?

Exercise 3. (Public Health Impact)

A philanthropic organization helped a town in Africa dig several wells to gain access to clean water. Before the wells were in place, an average of 120 infants contracted typhoid each month. After the wells were installed, the philanthropic organization surveyed for nine months and found an average of 90 infants contracted typhoid per month. Assume that the population standard deviation is 40 and the number of infants that contract typhoid is normally distributed.

1. Construct a 95% confidence Interval for the number of infants contracting typhoid after the wells were installed.
2. Conceptual question: how might the comparison between before and after the wells were dug be changed if the pre-well average came from a sample rather than knowing the population value?

Exercise 4. (Holiday Shopping)

A sample of holiday shoppers is taken randomly from a local mall to determine the average daily spending on gifts. From a preselected sample, the standard deviation was determined to be \$26. You would like to construct a 95% confidence interval for the mean daily spending on all holiday spending.

1. Find the appropriate sample size necessary to achieve a margin of error of \$5.
2. Find the appropriate sample size necessary to achieve a margin of error of \$8.

2 Hypothesis Testing Concepts

Exercise 1. (Understanding Significance Level)

If the chosen significance level is $\alpha = 0.05$, which of the following is correct?

- A. There is a 5% probability of rejecting a true null hypothesis
- B. There is a 5% probability of accepting a true null hypothesis
- C. There is a 5% probability of rejecting a false null hypothesis
- D. There is a 5% probability of accepting a false null hypothesis

Exercise 2. (Out-of-Wedlock Births)

It is generally believed that no more than 0.50 of all babies in a town are born out of wedlock. A politician claims that the proportion of babies born out of wedlock is increasing. Identify the correct null and alternative hypotheses.

- A. $H_0 : p = 0.50$ and $H_A : p \neq 0.50$
- B. $H_0 : p \leq 0.50$ and $H_A : p > 0.50$
- C. $H_0 : p \geq 0.50$ and $H_A : p > 0.50$
- D. $H_0 : p \geq 0.50$ and $H_A : p < 0.50$

Exercise 3. (LED Streetlights Impact)

Many cities are installing LED streetlights, in part to combat crime by improving visibility after dusk. An urban police department claims that the proportion of crimes committed after dusk will fall below the current level of 0.84 if LED streetlights are installed. Specify the null and alternative hypotheses.

- A. $H_0 : p = 0.84$ and $H_A : p \neq 0.84$
- B. $H_0 : p < 0.84$ and $H_A : p \geq 0.84$
- C. $H_0 : p \leq 0.84$ and $H_A : p > 0.84$
- D. $H_0 : p \geq 0.84$ and $H_A : p < 0.84$

Exercise 4. (One-Tailed Tests)

Which of the following are one-tailed tests?

- A. $H_0 : \mu \leq 10, H_A : \mu > 10$
- B. $H_0 : \mu = 10, H_A : \mu \neq 10$
- C. $H_0 : \mu \geq 400, H_A : \mu < 400$
- D. Both A and C

Exercise 5. (Two-Tailed Tests)

Which of the following are two-tailed tests?

- A. $H_0 : \mu \leq 10, H_A : \mu > 10$
- B. $H_0 : \mu = 10, H_A : \mu \neq 10$
- C. $H_0 : \mu \geq 400, H_A : \mu < 400$
- D. Both A and C

Exercise 6. (Significance Levels)

If the null hypothesis is rejected at a 1% significance level, then:

- A. The null hypothesis will be rejected at a 5% significance level
- B. The alternative hypothesis will be rejected at a 5% significance level
- C. The null hypothesis will not be rejected at a 5% significance level
- D. The alternative hypothesis will not be rejected at a 5% significance level

3 Applied Example

Exercise 1. (WHO Sugar Intake Study)

The World Health Organization (WHO) recommends that daily sugar intake does not exceed 50 grams for adults. A nutritionist checks the diet logs of 50 adults and finds that the average sugar intake is 60 grams. The nutritionist would like to determine if the average sugar intake is greater than the recommended limit. According to the WHO, the standard deviation of daily sugar intake for adults is 15 grams.

1. Calculate the value of the test statistic.
2. Find the p-value.
3. At confidence level $(1-\alpha) = 95\%$, can the nutritionist conclude that the average sugar intake is greater than the recommended limit? Represent the results as a bell-shaped graph. Mark clearly the mean, the test statistic, the rejection region, and the non-rejection region.